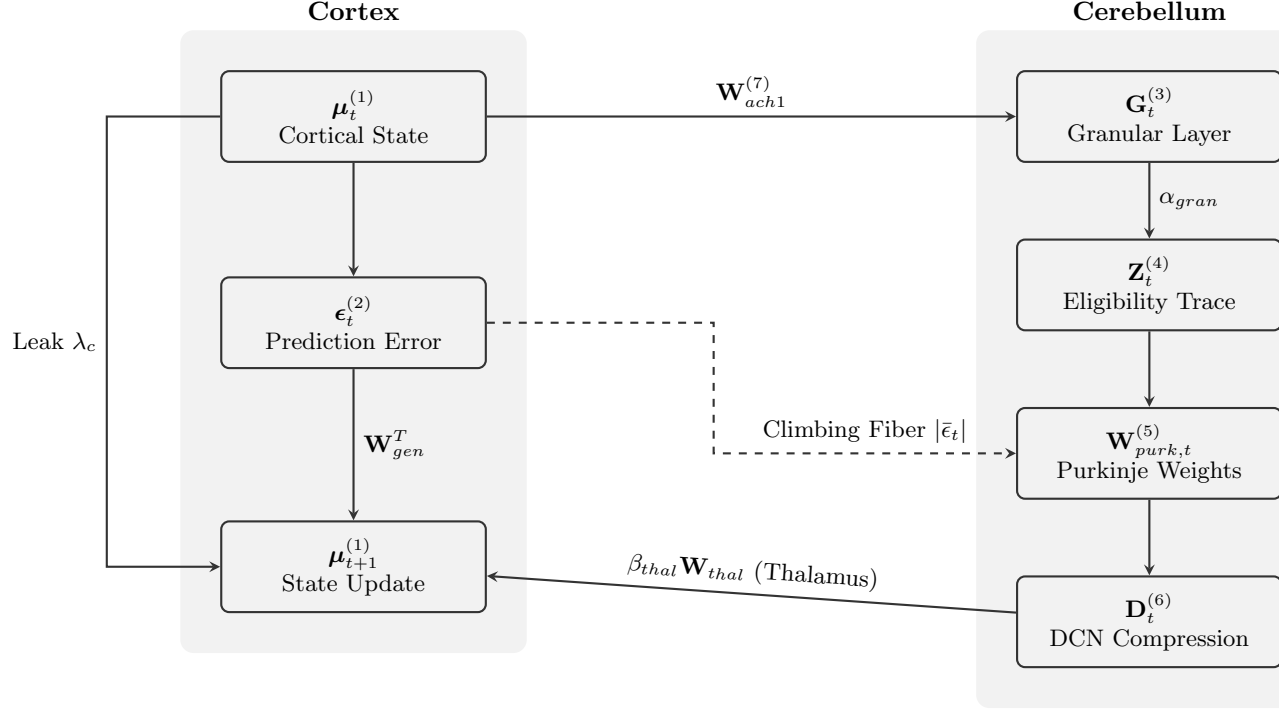




- (1) $\mu_{t+1} = \mu_t + \alpha_c [\mathbf{W}_{gen}^T \epsilon_t - \lambda_c ITI_t \mu_t + \beta_{thal} \mathbf{W}_{thal} \mathbf{D}_t]$
- (2) $\epsilon_t = \Pi \odot (\mathbf{I}_t - \hat{\mathbf{I}}_t)$
- (3) $\mathbf{G}_t = \mathbf{W}_{ach1} \mu_t$
- (4) $\mathbf{Z}_t = (1 - \beta_{gran}) \mathbf{Z}_{t-1} + \alpha_{gran} \mathbf{G}_t$
- (5) $\mathbf{W}_{purk,t} = \mathbf{W}_{purk,t-1} - \kappa(|\bar{\epsilon}_t|) \mathbf{Z}_t$
- (6) $\mathbf{D}_t = \mathbf{W}_{ach2} (\mathbf{G}_t \odot \mathbf{W}_{purk,t})$
- (7) $\mathbf{W}_{ach1}, \mathbf{W}_{ach2} \sim \mathcal{N}(0, \sigma_{ach}^2)$